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1. Differentiate circuit, packet and message switching.

→ circuit - circuit switching is a type of network configuration in which a physical path is obtained and dedicated to a single connection between two endpoints in the network for the duration of a dedicated connection.

packet - packet switching is the transfer of small pieces of data across various networks or in form of packets. These small pieces or packets allow for faster and more efficient data transfer.

message - Message switching is a network switching in which data is routed in its entirety from the source node to destination node, one hope at a time.

2. discuss the role of VCI.

→ A Virtual channel identifier distinguishes virtual channels created in a packet/cell switched network. A VCI has multiple circuits per communication channel and is primarily used for managing the unique identification of each created circuit.

3. discuss various types of framing.

→

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4. What is the length of the following?

(a). MAC address - 48 bits or 6 bytes

(b). IP address - 32 bits.

(c). Port address - 16 bit.

5. Draw and discuss the following.

(a). IP header - IP header is header information at the beginning of an Internet protocol packet. The header contains information about IP version, source IP address, etc.

(b). TCP header - TCP header is the first 24 ~~byte~~ bytes of a TCP segment that contains the parameters and state of an end-to-end TCP socket. It states the state of communication between two TCP endpoints.

(c). UDP header - UDP header is an ~~8 bit~~ 8-bytes fixed and simple header, while it is used for simple request-response communication when the size of data is less and hence there is lesser concern about flow and error control.

6. Write short notes on -

(a). HTTP - HTTP is a protocol for fetching resources such as HTML documents. It is the foundation of any data exchange on the Web and it is a client-server protocol, which means requests are initiated by the recipient, usually the Web browser.

(b). DNS - Domain Name system (DNS) is a naming database in which Internet domain names are located and translated into Internet Protocol (IP) addresses.

(c). FTP - File transfer protocol (FTP) is a way to download, upload, and transfer files from one location to another on the Internet and between computer systems.

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(42)

(d). TELNET- It is a network protocol used to virtually access a computer and to provide a two-way, collaborative and text-based communication channel between two machines.

(e). SMTP- Simple Mail transfer Protocol (SMTP) is used to send and receive email. It is sometimes paired with IMAP or POP3, which handles the retrieval of messages, while SMTP primarily sends messages to a server for forwarding.

7. An IPV4 packet has arrived with the first 8 bits as shown: 01000010. The receiver discards the packet. Why?

→ Min. header size of IPV4 = 20 byte

Max. header size of IPV4 = 60 byte

The first 4 bits indicate the ~~ver~~ version which is IPV4 (0100).

Now the next four bit indicates Header length which is $2 \times 4 = 8$ bytes.

which is not possible since the min. header length for IPV4 is 20 bytes.

Hence, receiver will reject the packet.

9.

→ MTU = 100 bytes.

size of IP header = 20 bytes.

So, size of data that can be transmitted in one fragment = 100 - 20
= 80 bytes.

size of data to be transmitted = size of datagram - size of header
= 1000 - 20
= 980 bytes.

Now, we have a datagram of size 1000 bytes.

So, $\frac{980}{80} = 13$ fragments.

④.

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(42)

10.

1 2 3 4 5 6 7 8 9 10 {Initially}
^

1 2 3 4 4 5 6 7 8 9 10 {Packet no°4 retransmitted}
^

1 2 3 4 4 5 6 7 7 8 9 10 {packet no° 10 retransmitted}
^

So, we transmitted packet number 4, 7, 10.

1 2 3 4 4 5 6 7 7 8 9 10 10 {final result}

Total count = 13.

11.

1 2 3 4 5 6 7 | 8 9 10 {packet no° 5 lost}
^ $

1 2 3 4 5 6 7 5 6 7 8 9 | 10
* ^ $

1 2 3 4 5 6 7 5 6 7 8 9 7 8 9 10
* ^ $

1 2 3 4 5 6 7 5 6 7 8 9 7 8 9 10 9 10

Total no° of transmissions = 18.

Sanchit Maheshwari
21155001 26

12.

Initial window size $\Rightarrow$ ~~C = 22 = 4 MSS.~~

C = 2 MSS.

for 2nd transmission $\rightarrow$ C = 22 = 4 MSS.

for 3rd transmission $\rightarrow$ C = 23 = 8 MSS.

When threshold is reached then additive increase.

for 4th transmission $\rightarrow$ C = 8 + 1 = 9 MSS

for 5th transmission $\rightarrow$ C = 9 + 1 = 10 MSS (fails)

window size reduced to $10/2$ = 5 MSS.

for 6th transmission $\rightarrow$ C = 1 MSS.

for 7th transmission $\rightarrow$ C = 21 = 2 MSS

for 8th transmission $\rightarrow$ C = 22 = 4 MSS

threshold reached now additive increase.

for 9th transmission $\rightarrow$ C = 4 + 1 = 5 MSS.

for 10th transmission $\rightarrow$ C = 5 + 1 = 6 MSS.

So, at the end of 10th successful transmission.

The congestion window size will be 6 + 1 = 7 MSS.

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(42)

⑥.

13.

CWND = 3000

effective window size = current window size - (last byte sent - last byte acked)

= 3000 - 2000
= 1000 bytes.

14.

Slow-start begins initially with a congestion window ~~will be~~ size of 1, 2, 4 or 10 MSS. The value of congestion window will be increased by one ~~with~~ with each acknowledgement received, effectively doubling the window size of each round-trip time.

The CWND increase by 1 MSS on every successful acknowledgements.

So, only statement (iv) is correct.

option [C] is true.

15.

Unacknowledged bytes = 10240 - 8192
= 2048 bytes.

Receiver available window size = 6KB - 2KB
= 4KB

Sender current ~~is~~ window size = min(4KB, 4KB) = 4KB

i.e. min(congestion window size, receiver available window size).

So, 4096 bytes.

16. The event at B which causes the sender to decrease its window is Triple duplicate ACK.